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Financial Time Series Forecasting Enriched with Textual Information

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***Abstract*—The ability to extract knowledge and forecast stock trends is crucial to mitigate investors' risks and uncertainties in the market. The stock trend is affected by non-linearity, complexity, noise, and especially the surrounding news. External factors such as daily news became one of the investors' primary resources for buying or selling assets. However, this kind of information appears very fast. There are thousands of news generated by different web sources, taking a long time to analyze them, causing significant losses for investors due to late decisions. Although recent contextual language models have transformed the area of natural language processing, models to make predictions using news that influence stock values still face barriers such as unlabeled data and class imbalance. This paper proposes a hybrid methodology that enriches the time series forecasting considering textual knowledge extracted from sites without a widely annotated corpus. We show that the proposed method can improve forecasting using an empirical evaluation of Bitcoin prices prediction.**

***Index Terms*—Time Series, Text Mining, Forecasting, Bitcoin**

I. Introduction

Financial markets perform an essential role in modern society's economy and organization, influencing several countries' economic development. This relevance attracts researchers' and business professionals' attention for their theoretical skills and practical applications [1].

The stock market can be seen as a complex system influenced by many aspects, such as socio-political events, news about investor behaviors, companies' partnerships, and control organizations [2]. Information about these aspects became one of the primary resources to support investors' decisions about buying or selling. However, this kind of information appears very quickly. There are thousands of news continuously generated by different resources, taking a long time to analyze it one by one, which can cause significant losses to investors or small businesses due to late decisions. So, we have two types of problems: a large amount of news and the time it takes to analyze these news' impact on the market.

In recent years, several approaches have been proposed to solve the problem of forecasting financial time series using three main approaches [1]: statistical methods, machine learning-based, and text mining-based models. Statistical methods assume that the data follows a known a priori distribution that is used as a parameter to build the forecast model. Machine learning-based models may use Neural Networks, Support Vector Machines, Nearest Neighbors, among other supervised learning [3], with a recent increase of methods based on attention mechanisms [4]–[6]. Using time series historical data to induce a model is a common factor between these approaches.

Researchers have tried to predict future market movements based just on text mining [7], [8], [9], [10]. They use financial news, reports, or even social networks to extract qualitative information from companies and use it to predict the price. Recent studies show that we can improve time series forecasting by incorporating additional information from the surroundings to time series analysis [11], [12], [13].

Only a few studies use this external knowledge extracted by a text mining model to improve the model-based performance. Therefore, we propose a new hybrid methodology to enrich the time series with external information.

Our proposal is divided into two main parts. First, the data mining model performs a fine-tuning of the pre-trained Bidirectional Encoder Representations from Transformers (BERT) for sentiment classification without needing a fully manually labeled corpus. For that, we used a little annotated corpus followed by a Label Propagation algorithm to propagate the label to unlabeled texts using BERT embedding representations. In the second part, we learned the most relevant and abstract features from a list of technical indicators in an unsupervised manner using AutoEncoders. Then, these features and textual information are used as input to forecast the price.

The experimental results indicate that the incorporation of news is potentially useful for improving the accuracy and risk management in the forecast and allowing to analyze and interpret at the semantic level the external factors involved in the time series analysis. With an experiment on Bitcoin assets, we demonstrate that enriching the time series forecast with textual information presents an increase of 3.7% of R^2 and reduces the mean absolute percentage error by 1.62%.

II. Related Work

Statistical forecasting models, such as Autoregressive (AR), Autoregressive Integrated Moving Average (ARIMA), and Moving Average (MA), were considered state-of-the-art to model time series and forecast for decades [2]. Usually, these algorithms are parametric, given that they require a priori knowledge of the model's parameters [3]. Once most of them assume that the time series is generated from a linear model, non-linear patterns cannot be captured by these models [14].

In contrast to statistical methods, Machine Learning (ML) models, such as Neural Networks and Support Vector Ma-

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chines, can capture non-linear and non-stationary relationships, describing the data's characteristics without a priori knowledge about the data distribution [1]. Over the past few years, there has been an increase in methods based on ML. The literature had used models based on Deep Learning (DL), such as the Long Short Term Memory (LSTM) [15], [16]. Other authors mixed LSTM with unsupervised learning, specifically with AutoEncoders (AE), to learn a reduced representation of the multivariate time series before predicting future values [17]. More recently, other authors have used attention mechanisms for forecasting [4], [6], [18]. These models contain an encoder to extract the most relevant characteristics of the series. Another step of the pipeline contains a temporal attention mechanism to select the most relevant hidden states.

Besides, Natural Language Processing (NLP) can be used to aid the time series forecast. For instance, sentiment analysis and opinion mining [19] help understand customers' and investors' positions regarding a company, a product, political events, or a specific stock. [10] incorporates a sentiment indicator of the result of the processing of tweets and financial texts to improve asset prediction. [9] analyze cryptocurrency forums with higher capitalization. [11] propose to enrich the time series forecasting from the semantic web perspective using LSTM and Convolutional Neural Networks (CNN). All these works illustrate the positive impacts of semantic knowledge on time series.

III. Fundamental Concepts

Before introducing our proposal, we present basic concepts on time series, text embeddings, and Label Propagation, which are necessary to comprehend our method.

A. Time Series

Time series is a sequence of observations chronologically ordered. Formally, a time series can be stated as an ordered set of values Z of size m, $Z = (z_1, z_2, z_3...z_m)$ where $z_t \in \mathbb{R}$ for all $t \in [1, m]$. Each z_t value in Z is an observation of the time series at time t. A time series can be deterministic or stochastic [20]. Suppose a mathematical function synthesizes the series values in terms of $y = f(time)$. In that case, the series is called deterministic, i.e., it has a regular and predictable behavior. When the series contains a random term ϵ, $y = f(time, \epsilon)$, the series is stochastic or nondeterministic.

B. Word Embeddings

The primary purpose of word embeddings is to project words in a continuous vector space. In this space, semantically or syntactically related words should be located in the same region. Algorithms to induce this space are usually based on the distributional hypothesis, which considers that words with similar meanings tend to appear in similar contexts [21].

The embeddings are generally learned in an unsupervised manner from a substantial unlabeled corpus. After computationally intense training, the authors of that embedding usually make it freely available so that other researchers may use the learned word space in different NLP tasks.

[22] made a revolutionary change in the field of NLP by introducing Bidirectional Encoder Representations from Transformers (BERT). It uses a deep bidirectional architecture based on Transformers to produce pre-trained context-dependent embeddings. BERT has the advantage of considering the context within which the word appears.

Aside from capturing noticeable differences like polysemy, the context-informed word embeddings capture other forms of information that result in more accurate feature representations, resulting in better model performance.

C. Label Propagation

In many application domains, labeling data is expensive, especially with the increasing volume of data. For this reason, many real-world data only have labels for a small subset of their data points [23]. Nevertheless, the labeled examples are essential for supervised tasks, such as classification and regression.

In this scenario, we may rely on transductive learning algorithms. For example, label Propagation (LP) [23] is a transductive learning algorithm that considers that similar data tend to have the same class label. Based on this assumption, LP propagates labels from initially labeled examples to all the other (unlabeled) data points according to a network defined by a proximity measure.

IV. Proposed Method

We propose a hybrid approach that allows incorporating external information into the time series forecasting model to reduce the prediction error. The study is done on *Bitcoin*, a digital asset in the crypto market. Figure 1 summarizes the proposed method.

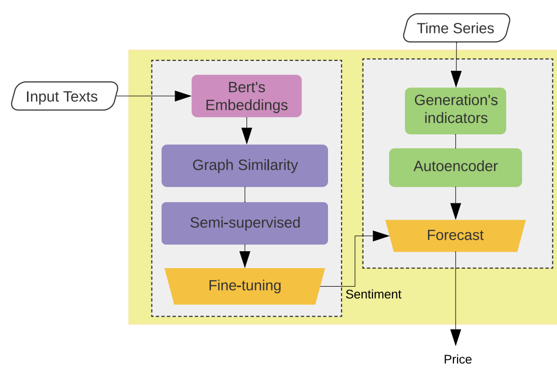

Fig. 1. Proposed methodology

First, on the left, we develop the text mining pipeline. It uses pre-trained contextual embeddings to represent each text in the continuous space. Next, we build a similarity network using distances calculated in this transformed space. Some of the texts in the network are labeled with their sentiment, but most of them have no label associated. The similarity network serves as the input for the semi-supervised algorithm that propagates the label to other neighboring nodes according to their proximity. We perform fine-tuning of the pre-trained

BERT to find a new embedding space that fits our data and objective (sentiment classification) better in the last step.

The second pipeline, on the right, refers to the time series analysis model. We generated a dataset of technical indicators representing the AutoEncoder (AE) input. The AE learns a compact representation of the input features in an unsupervised procedure by encoding it to reduced space and recovering the original values.

We follow the forecasting using the compact representation learned by the AE concatenated with the sentiment predicted by the text mining pipeline.

As we used different texts representations to calculate the distances between the news to create the similarity network and different fine-tuned language models to perform the sentiment analysis, we present the applied techniques next.

- **Feature Representation**. There exist different ways to represent a text in a structured and numerical way. In this work, we used a word representation based on BERT Embeddings. We generate one embedding for each word of the given text, and, as a result, we get a sequence of embeddings with a fixed size for a text.
- **BERT fine-tuned** We used transfer learning to fine-tune BERT to our specific task of predicting market sentiment from financial Bitcoin news. We adapt BERT to our target task using a small corpus. First, we re-use the model parameters, layers, and weights of the pre-trained model as the initial configuration. Then, by adding a softmax function to BERT, we can fine-tune it by training it to predict the sentiment-labeled texts from our corpus. We use DistilBERT [24], a distilled version of BERT, as part of our experimental results. DistilBERT reduces model's size by 40%, maintaining 97% of its capabilities.

V. Experimental Evaluation

This section presents how we approach each of the steps illustrated in Figure 1, from the data acquisition to the fusion of text and time series pipelines. For the sake of reproducibility, we make available all the source codes and detailed results on this paper's GitHub repository[1].

A. Dataset Description

As no organized and freely available dataset fits our proposal, we collected two parallel datasets: a corpus of news and time series from indicators related to Bitcoin assets.

The corpus is composed of news collected from significant websites in the financial information ecosystem[2]. A problem detected after data collection is that crawling these websites, even guided by the keyword "Bitcoin", returns hundreds of news that do not bring relevant information to the studied phenomenon. On the other hand, our search returned several relevant texts. For instance, news on decisions regarding concepts like Security Exchange Commission (SEC), Exchange Trade Funds (ETF's), and Chicago Board Options Exchange (CBoE) may influence the prices of the target cryptocurrency.

[1]Omitted for review purposes.

[2]Sites sources: cointelegraph.com, coindesk.com, newsbtc.com, ccn.com, decrypt.com, ambcrypto.com, cryptodaily.co.uk, ccn.com

We applied a second filter on the retrieved texts, including or excluding news according to a list of interest keywords, presented in Table I, to avoid irrelevant news and a significantly imbalanced dataset. Finally, our corpus contains $101, 11$ news from 1 November 2018 to 3 November 2020, being that all news contain words in the "included" list and no words in the list of "excluded" terms.

TABLE I
Keywords Included and Excluded

Action	Keyword
Excluded	'price analysis', 'cryptocurrency price analysis', 'bitcoin cash', 'cryptocurrency price review', 'cryptocurrency marketcap','bitcoin sv', 'ripple', 'bitcoin gold', 'bitcoin cash', 'xrp', 'bch', 'tron', 'ethereum', 'eos', 'neo', 'eth'
Included	'bitcoin', 'blockchain', 'bakkt', 'vaneck', 'vaneck bitcoin etf', 'solidx', 'bitcoin futures', 'cryptocurrency', 'sec', 'cboe', 'bitcoin trust', 'cme', 'distributed ledger', 'securities exchange commission', 'halving' 'etf', 'cme group', 'blockchain leader', 'coronavirus'

The second dataset concerns the time series of historical Bitcoin prices. The dataset was obtained from a web platform, which offers public access from *Bitfinex*[3], which have historical trading data with windows from 1 minute to 1 day in *OHCL* format (Open, High, Close, Low). In our experiments, we opted for a 4 hours time window. From the OHCL, we extracted technical indicators commonly used in Technical analysis [25], listed in Table II.

TABLE II
List of technical indicators.

Technical Indicators	Description
Open /Close price	Daily open and close price
High /Low price	Daily Highest/Closest price
Volume	Daily Volume
MACD	Moving Average Convergence Divergence
RSI/Stochastic RSI	Relative Strength Index, Stochastic RSI
EMA20	Exponencial Moving Average 20 periods
MAE5/MAE10/MAE200	Moving Average 5, 10, 200 periods
WMA20	Weighted Moving Average para 20 periods
ROC	Price Rate of Change
SMI	Stochastic Momentum Index
WDAD	William Variabel Acumulation Distribution
CCI	Commodity Channel Index
BBANDS	Bolling Bands

B. Critical News Identification

This step aims to build the initial short dataset of labeled texts used in the Label Propagation step. We adopt the trend of time series as support for labeling. We assume that reports related to the cause of (or are caused by) price changes coincide with time windows with up or down trends. However, not every news in the adjacent time window of non-, down, or uptrends is necessarily linked to this event.

To identify the most relevant news for Bitcoin, we used a strategy that consists of two steps: (i) identifying the trend periods with high volatility in all historical time series and (ii) selecting only the most important news from these trend

[3]https://www.bitfinex.com/

periods. We used the idea from [26] with some changes. We calculate the returns *open-to-close* together with value of the *Relative Strength Index* (RSI) indicator to identify only trend periods. RSI is a momentum indicator used in technical analysis that measures the magnitude of recent price changes to evaluate overbought and oversold [25]. Equation 1 formally defines this idea, illustrated in Figure 2.

$$\begin{aligned} r_t &= O_{t+f} - C_t \\ rsi_t &= RSI_{t+f} \\ \text{Volatil Dates} &= \begin{cases} up, \quad if \quad r_t > 0, rsi > rsi.high \\ down, if \quad r_t < 0, rsi \leq rsi.low \end{cases} \end{aligned} \tag{1}$$

Fig. 2. Labeling short dataset using the time series historial.

To calculate the returns r_t, we use the time series's opening price O_{t+f} in time t. We empirically defined the best values for f as 5 and 7. C_t is the closing price, and rsi_t is the RSI value in the time t, which measures the speed and trend of price movement. RSI is configured with the maximum and minimum values (*rsi.high* and *rsi.low*) of 75 and 25, respectively.

After identifying the up or down trend dates, we filtered the most relevant news in these trends. We defined important news based on [13], as the news with at least 2 texts similarity equal published in the same trend periods within their k-similarity news neighborhood. The similarity is calculated using the cosine distance between two *n*-dimensional vectors, consider two documents represented by their (BERT) embeddings.

C. Label Propagation

We created an undirected graph, where each node represents a news document, where an edge occurs if: (i) the minimum similarity between the nodes is 0.85 and (ii) each node has at most 3 neighbors. The graph contains 10111 nodes and 30334 edges.

We compare two regularization techniques: Gaussian Fields and Harmonic Functions (GFHF) and Learning with Local and Global Consistency (LLGC) [27]. The first one is based on GFHF, which does not change the labeled data. Besides, we use the regularization framework LLGC, which allows changing the initial labels under certain circumstances since they may have been mislabeled or are noisy.

D. Fine tuning Configuration

We use two variations of the context-aware model language: BERT and DistilBERT. We fine-tuned these models to our domain with the described corpus annotated with three classes: *up-trend*, *down-trend*, and *non-trend*. We used the implementations of *bert-base-uncased* and *distilbert-base-uncased* from Hugging Face[4]. For all models, we used a maximum sequence length of 512, batch size of 16 examples, 3 training epochs, and learning rate of $5e-4$.

E. Abstract Features Representations

The purpose of this phase of our proposal is to find relevant transformed features from the financial time series from the list of indicators shown in Table II. These new characteristics are learned in an unsupervised manner, using an AutoEncoder (AE).

The AE training process consists of two parts: *encoding* and *decoding*. The *encoding* process maps input data x to an abstract (and reduced) representation of features in a hidden layer. The *decoding* step reconstructs the original data from the hidden layer.

Our AE architecture is formed by a series of stacked ReLu and linear layers that allow learning the relationships between input features. The encoder contains: $Input \rightarrow ReLu \rightarrow Linear \rightarrow ReLu \rightarrow Linear \rightarrow ReLu \rightarrow$ Encoder Layer. Then, the decoder mirrors these layers to reconstruct the input. Through the dimensionality reductions, we encoded 25 features to an encoder layer with four new abstract features.

F. Enriching Time series

Once we have defined each step of both pipelines, we need an aligning strategy to enrich the time series with textual data. Specifically, each time window represented by a time series of observations may have one, many, or no text temporarily associated with it. The proposed strategy is based on concatenating the weighted average news labeled by the LP algorithm to the time series features.

First, recognizing that the market operates 24 hours a day, 7 days a week, we define a daily transaction that starts at 21:00:00 hrs until 21:00:00 hrs of the next day. News will be aligned to each transaction according to their publication dates. As we set windows of 4 hours, a daily transaction comprises 6 observations of the time series.

Given the news classification in the transaction day i, we aggregate it to a single value by a weighted average. Equation 2 defines how to calculate this score.

$$Tr_i = argmax_{k \in \{+,-,0\}} \left(\frac{\sum N^k}{N_T} \times \alpha[k]] \right) \tag{2}$$

where Tr_i represents the frequency of each class. It is calculated by counting the total number of news for one class and dividing by the total number of news N_T that happened in a transaction i during a day. $\sum N^k$ represents the number of examples classified in the k-th class label. α is a vector of weights that allows to define different importance values

[4]https://huggingface.co/

to each class. Since not all news have the equal importance – mainly because there are thousands of *non trend* news –, we set the weight $\alpha = 0.1$ to the *no trend* (0) class, and *up trend* (+) and *down trend* (-) have a higher weight: $\alpha = 0.9$.

VI. Results and Discussion

In this section, we present the outcomes of each step of our proposal. After performing the identification of critical news steps mentioned in section V-B, we obtain a corpus of news annotated regarding the time series trend. Table III shows the number of examples labeled in each class and the number of unlabeled texts in this intermediate corpus.

TABLE III
Initial Labeled news by Critical Identification method.

	Up-trend	Down-trend	Non-trend	Unlabeled
# of documents	80	135	5979	3917

Next, we construct the similarity graph and apply the Label Propagation to obtain more labels in our corpus. Besides, we compared LP with another strategy, using the same graph topology as input for Node2Vec [28] algorithm, which learns graph embeddings in a lower dimension. The parameter setting used for Node2Vec with 3 dimensions, 10 paths per node, length of each node 50, $p = 1$, and $q = 1$. With this new vector in the lower dimension as input, we applied the K-means to cluster the examples. We fixed $k = 3$ as cluster numbers.

Table IV show the results obtained in this step. For this experiment, we use 30% of the labeled examples as initially labeled nodes and measure the accuracy in the remaining 70%.

TABLE IV
Label Propagation comparison GFHF and LLGC. The dimension of both Word2Vec and BERT is 512.

Representation	Up	Down	Non	Accuracy
LP + GFHF (iter=50)				
TF-IDF	87	158	9790	0.940
Word2Vec based	82	146	9883	0.963
BERT embedding	84	146	9881	0.962
LP + LLGC ($\mu = 0.85$, iter=50)				
TF-IDF	87	135	9828	0.946
Word2Vec based	82	135	9886	0.965
BERT embedding	83	135	9887	0.965
Node2Vec (p=q=1) + Kmeans				
TF-IDF	2197	2720	5102	0.540
Word2Vec based	1892	3286	4933	0.951
BERT embedding	3605	3275	3231	0.963

We observe that LLGC and GFHF frameworks perform similarly. It may occur because the initial dataset was correctly labeled, so LLGC did not change many initial labels. From now on, we use the LLGC with $\mu = 0.85$ as LP technique and keep the Node2Vec for comparison purposes.

With the previously labeled corpus, we analyze the classification accuracy obtained by different models using strategies for labeling. For comparison purposes, we experimented with the fine-tuned BERT and DistilBERT, as well as with NB-SVM [29] and FastText [30]. Table V presents the results. In this case, we use 80% of the data for training and the remaining for test.

TABLE V
Metrics for different classifier using labeling strategies.

	NB-SVM	FastText	BERT	DistilBERT
Labeled: Label Propagation				
Accuracy	88.49	93.00	97.00	**98.00**
Precision	39.38	**48.00**	32.00	33.00
Recall	**40.95**	37.00	33.00	33.00
F1-Score	**39.83**	39.00	33.00	33.00
Labeled: Node2Vec+KMeans				
Accuracy	78.00	79.00	**85.00**	**87.00**
Precision	79.00	79.00	**85.00**	**87.00**
Recall	78.00	80.00	85.00	**87.00**
F1-Score	79.00	79.00	**85.00**	**87.00**

In general, the performance of BERT and DistilBERT using the Label Propagation strategies was inferior. While they achieve the best accuracies, we can notice that it concerns the class imbalance problem due to the fact they achieved 33% or less in *precision*, *recall*, and *f1-score*. BERT and DistilBERT have not been capable of handling skewed classes, mostly learning the majority class.

The unfortunate results of LP indicates that BERT does not generalize well when the classes are unbalanced, and we would need to make additional efforts.

Finally, we present the results of the time series forecasting enriched with textual data. We measure the effectiveness of various forecast methods, we consider four different evaluation metrics: R-square (R^2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

To demonstrate the reduction of error, we added the sentiment value calculated in Equation 2. We set $\alpha[+] = 0.9$, $\alpha[-] = 0.9$ and $\alpha[0] = 0.1$ values for up, down, and non trend classes, respectively. We use LSTM to model one single feature, the price observations as input, as baseline. The LSTM-ENC (AutoEncoder) model learns from the extracted features explained in V-E. We perform the time series prediction using 4 abstract features.

We also compare our models with a Double Attention-Recurrent Neural Network proposed (DA-RNN) [4]. This model uses two attention mechanisms internally: (i) one encoder that can select the relevant driving series and (ii) the decoder, which selects the relevant encoder hidden across all time steps. We experimented with the DA-RNN with and without the predicted sentiment.

Table VI presents the results of the forecasting phase.

TABLE VI
Comparative results forecast error of the time series enriched with sentiment

Model	Sentiment	R^2	MAE	RMSE	MAPE
LSTM	No	0.934	0.195	0.257	34.10
LSTM-ENC	No	0.900	0.072	0.098	6.934
LSTM-ENC-SE	Yes	0.777	0.034	0.043	7.13
DA-RNN [4]	No	0.945	0.016	0.021	3.622
DA-RNN-SE	Yes	**0.982**	**0.009**	0.0003	**2.006**
DA-RNN-N2V-SE	Yes	0.978	0.010	**0.0002**	2.166

We can observe that the error of LSTM is considerably higher than LSTM-ENC. It possibly happens because the LSTM only considerates the close price and ignores other features or technical indicators. When we enrich the LSTM-ENC (we refer to the models enriched with sentiment with "SE"), we observe a reduction of error and R^2.

On the other hand, the market sentiment analysis's influence is more apparent for the DA-based models. Using the sentiment causes a reduction in all error measures and an increase in R^2 of 3.7%. Although the Node2Vec approach obtained better results in the previous steps, we noticed that it did not differ from the same forecaster using the LP approach's sentiment prediction. It may occur because, although the second approach leads to a lower precision in the classification, the scores for each class obtained by DistilBERT are better distributed.

VII. Conclusion

We presented a hybrid method that allowed us to reduce financial time series forecasting error when we incorporated the sentiment analysis of related news into the time series model. Although this indicates that external knowledge can be helpful, limitations became clear in our experiments. Therefore, we use these limitations as guidelines for stating future work.

For better comprehension and deeper evaluation of the text analysis, it is essential to have a more detailed info of the task Named Entity Recognition (NER) to know which entity is talking about and which entity an action is done. Consider, for instance, that company A has partnered with company B. Without NER analysis, we do not know which entity the impact happens, consequently which time series will be enriched with company A or B.

Although we used real news associated with an asset to improve the forecasting, our proposal was not designed to operate on a platform. So, we intend to embed our approach in a buy-hold-sell mechanism and assess the return of investment.

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